

Estimation and Fermi Problems

Mission

Warm-up and worked example

Name: _____

Date: _____

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YOUR MISSION Make reasonable assumptions, estimate quantities, and give useful bounds.

SMART MOVE Draw a simpler case and organize what you know.

Math Talk

About how many steps does it take to cross a room? How could you estimate?

My idea:

New Words

represent	deduce	assumption
My meaning or picture:	My meaning or picture:	My meaning or picture:

Worked Example

Do this example with your teacher. Say what changes and what stays the same.

1. Ask: How many books could fit on one shelf?
2. List needed information.

What the example shows: _____

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Learn it and show your thinking

Name: _____

Date: _____

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YOUR MISSION Follow the learning path, then explain the big idea.

SMART MOVE Draw before asking for a rule.

[] **Step 1:** Ask: How many books could fit on one shelf?

[] **Step 2:** List needed information.

[] **Step 3:** Choose reasonable measurements.

[] **Step 4:** Calculate an estimate.

[] **Step 5:** State a lower and upper bound.

[] **Step 6:** Test the estimate if practical.

My model, drawing, or organized work:

The big idea in my own words:

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Practice lab

Name: _____

Date: _____

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YOUR MISSION Use a model, equation, or explanation for every task.

SMART MOVE Check whether your answer is reasonable.

1. Number of tiles on a floor.

2. Number of cups in a container.

3. Number of pages read in a month.

4. Number of minutes spent eating breakfast in a year.

Choose one answer and prove it another way:

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Challenge and final check

Name: _____

Date: _____

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YOUR MISSION Try an unfamiliar problem and defend your reasoning.

SMART MOVE A wrong first idea can still lead to a strong solution.

Challenge Mission

Estimate how many pencils would be needed to stretch across the length of the home. Defend the assumptions.

My plan:

My solution and proof:

Final Check

Explain why a reasonable estimate can be valuable even when it is not exact.

☐ I can teach it

☐ I need practice

☐ I need help

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Reusable math tool

Name: _____

Date: _____

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YOUR MISSION Use this page to draw, model, organize, and check.

SMART MOVE Draw a simpler case and organize what you know.

Problem-Solving Organizer

What I know

What I need to find

My model or table

My equations

My check and explanation
